

Ex. 2. What value will replace the question mark in each of the following questions?

(i)? - 1936248 = 1635773 (ii) 9587 - ? = 7429 - 4358

Sol. (i) Let $x - 1936248 = 1635773$. Then, $x = 1635773 + 1936248 = 3572021$.

(ii) Let $9587 - x = 7429 - 4358$.

Then, $9587 - x = 3071 \Rightarrow x = 9587 - 3071 = 6516$.

Ex. 3. What could be the maximum value of Q in the following equation?

$$5P9 + 3R7 + 2Q8 = 1114$$

Sol. We may analyse the given equation as shown:

Clearly, $2 + P + R + Q = 11$.

So, the maximum value of Q can be $(11 - 2)$, i.e. 9 (when $P = 0$, $R = 0$).

Ex. 4. Simplify: (i) 5793405×9999 (ii) 839478×625

Sol. (i) $5793405 \times 9999 = 5793405 \times (10000 - 1) = 57934050000 - 5793405 = 57928256595$.

(ii) $839478 \times 625 = 839478 \times 5^4 = 839478 \times \left(\frac{10}{2}\right)^4 = \frac{839478 \times 10^4}{2^4} = \frac{8394780000}{16} = 524673750$.

Ex. 5. Evaluate: (i) $986 \times 137 + 986 \times 863$ (ii) $983 \times 207 - 983 \times 107$

Sol. (i) $986 \times 137 + 986 \times 863 = 986 \times (137 + 863) = 986 \times 1000 = 986000$.

(ii) $983 \times 207 - 983 \times 107 = 983 \times (207 - 107) = 983 \times 100 = 98300$.

Ex. 6. Simplify: (i) 1605×1605 (ii) 1398×1398

Sol. (i) $1605 \times 1605 = (1605)^2 = (1600 + 5)^2 = (1600)^2 + 5^2 + 2 \times 1600 \times 5$
 $= 2560000 + 25 + 16000 = 2576025$.

(ii) $1398 \times 1398 = (1398)^2 = (1400 - 2)^2 = (1400)^2 + 2^2 - 2 \times 1400 \times 2$
 $= 1960000 + 4 - 5600 = 1954404$.

Ex. 7. Evaluate: (i) $475 \times 475 + 125 \times 125$ (ii) $796 \times 796 - 204 \times 204$

Sol. (i) We have $(a^2 + b^2) = \frac{1}{2}[(a+b)^2 + (a-b)^2]$

$$\therefore (475)^2 + (125)^2 = \frac{1}{2}[(475+125)^2 + (475-125)^2] = \frac{1}{2}[(600)^2 + (350)^2]$$

$$= \frac{1}{2}[360000 + 122500] = \frac{1}{2} \times 482500 = 241250$$

(ii) $796 \times 796 - 204 \times 204 = (796)^2 - (204)^2 = (796 + 204)(796 - 204)$
 $= (1000 \times 592) = 592000$.

$$[\because (a-b)^2 = (a+b)(a-b)]$$

Ex. 8. Simplify: (i) $(387 \times 387 + 113 \times 113 + 2 \times 387 \times 113)$

(ii) $(87 \times 87 + 61 \times 61 - 2 \times 87 \times 61)$

Sol. (i) Given Exp. $= (387)^2 + (113)^2 + 2 \times 387 \times 113 = (a^2 + b^2 + 2ab)$, where $a = 387$ and $b = 113$
 $= (a + b)^2 = (387 + 113)^2 = (500)^2 = 250000$.

(ii) Given Exp. $= (87)^2 + (61)^2 - 2 \times 87 \times 61 = (a^2 + b^2 - 2ab)$, where $a = 87$ and $b = 61$
 $= (a - b)^2 = (87 - 61)^2 = (26)^2 = (20 + 6)^2 = (20)^2 + 6^2 + 2 \times 20 \times 6 = (400 + 36 + 240)$
 $= (436 + 240) = 676$.

Ex. 9. Find the square root of $4a^2 + b^2 + c^2 + 4ab - 2bc - 4ac$.

(Campus Recruitment, 2010)

Sol. $\sqrt{4a^2 + b^2 + c^2 + 4ab - 2bc - 4ac} = \sqrt{(2a)^2 + b^2 + (-c)^2 + 2 \times 2a \times b + 2 \times b \times (-c) + 2 \times (2a) \times (-c)}$
 $= \sqrt{(2a + b - c)^2} = (2a + b - c)$.

Ex. 10. A is counting the numbers from 1 to 31 and B from 31 to 1. A is counting the odd numbers only. The speed of both is the same. What will be the number which will be pronounced by A and B together?

(Campus Recruitment, 2010)

Sol. The numbers pronounced by A and B in order are:

A	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31
B	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Clearly both A and B pronounce the number 21 together.

$$\begin{array}{r} \textcircled{1} \textcircled{2} \\ 5 \ P \ 9 \\ 3 \ R \ 7 \\ \hline 2 \ Q \ 8 \\ \hline 1 \ 1 \ 1 \ 4 \end{array}$$